BIBI ZAINAB

Phone: +91 861 824 0798 | Email: zk946190@gmail.com
LinkedIn: linkedin.com/in/bibi-zainab-dhalayat-929b2b41a | GitHub: github.com/zk946190-ui
Portfolio: github.com/zk946190-ui/zk946190-ui

ECE GRADUATE | EMBEDDED SYSTEMS | PCB DESIGN | PYTHON & ML

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with project experience in both hardware and software. On the hardware side: Arduino-based embedded systems, sensor interfacing and PCB design training in KiCad. On the software side: Python machine learning, an LLM-based AI agent and cybersecurity, including a cybersecurity internship. Seeking an entry-level role in PCB design, embedded systems or electronics hardware, or in Python, machine learning or cybersecurity.

TECHNICAL SKILLS

- **PCB Design & Embedded:** KiCad, schematic capture, PCB layout fundamentals; Arduino Mega, Embedded C/C++, sensor and actuator interfacing (load cell with HX711, inductive sensor, servo motor, I2C LCD, keypad), UART modules (GSM, Bluetooth HC-05), EEPROM
- **VLSI Fundamentals:** Verilog HDL basics, RTL concepts, introductory FPGA and ASIC concepts
- **Python & ML:** Python, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Flask, LLM API integration (Groq)
- **Cybersecurity & IT:** Networking fundamentals (TCP/IP, DNS, subnetting), Kali Linux, VMware, Windows Server, Active Directory, AWS EC2, Apache, Nginx, PowerShell, Bash
- **Tools:** Git/GitHub, Linux, Arduino IDE, VS Code

PROJECTS

Reverse Vending Machine (Team project, 4 members)
github.com/zk946190-ui/reverse-vending-machine
Arduino Mega, Embedded C/C++, HX711 load cell, inductive sensor, GSM 4G module, HC-05 Bluetooth, I2C LCD

- Built a campus recycling machine that sorts plastic, metal and glass bottles using an inductive sensor, an HX711 load cell and two servo motors; tested with 50 bottles per type: 90% (plastic), 100% (metal), 80% (glass) sorted correctly.
- Implemented keypad USN login, reward points stored in EEPROM, a Bluetooth admin panel (ADD USER / DELETE USER / LIST) and GSM SMS notifications, including a "system full" alert.

Course Recommendation Agent
github.com/zk946190-ui/AI-agent
Python, Groq LLM API, JSON

- Built an AI agent that reads student profiles and a course catalogue (JSON) and uses the Groq LLM to generate a personalised learning path, prerequisites, timeline and career advice.

AI-Based Phishing URL Detection
github.com/zk946190-ui/phishing-url-detection
Python, Scikit-learn, Pandas, TF-IDF, Random Forest, Flask

- Trained a TF-IDF + Random Forest classifier to label URLs as phishing or safe (78.1% accuracy on 498 test URLs) and built a Flask web dashboard for real-time checks.

Skin Cancer Detection using CNN
github.com/zk946190-ui/skin-cancer-detection-cnn
Python, TensorFlow/Keras, EfficientNetB7

- Classified skin lesion images as benign or malignant; improved test accuracy from 45.5% (CNN from scratch) to 83.3% using EfficientNetB7 transfer learning.

INTERNSHIP

Cybersecurity Intern — GTTC, Hubli | February 2026 – May 2026

- Studied networking fundamentals (IP addressing, subnetting, ports, protocols, OSI model) and set up Kali Linux, Windows Server and Ubuntu in VMware.
- Managed users, groups and organizational units in Active Directory; launched AWS EC2 servers and configured Apache and Nginx (hosting, reverse proxy, load balancing).
- Practised PowerShell and Bash scripting; completed a capstone project, the phishing URL detector above.

EDUCATION

B.E. in Electronics and Communication Engineering | 2022–2026
Government Engineering College, Talakal, Koppal (Visvesvaraya Technological University, Belagavi)
CGPA: 7.95/10

TRAINING & CERTIFICATIONS

- PCB Design Training — KiCad (schematic capture and PCB layout fundamentals)
- Cybersecurity and Ethical Hacking — Cisco learning certificates
- Cybersecurity internship completion certificate — GTTC, Hubli